

Where AI Is Creating Measurable Economic Value

in B2B Services, Manufacturing, Industrial Services and Distribution — and what a private-equity investor should do differently

The central analytical chain of this report is **adoption** → **operational change** → **economic impact** → **financial impact** → **competitive dynamics** → **investment implication**. Every quantitative claim is sourced, dated and attributed to a geography. Where reliable evidence does not exist, the report says so rather than substituting a plausible number.

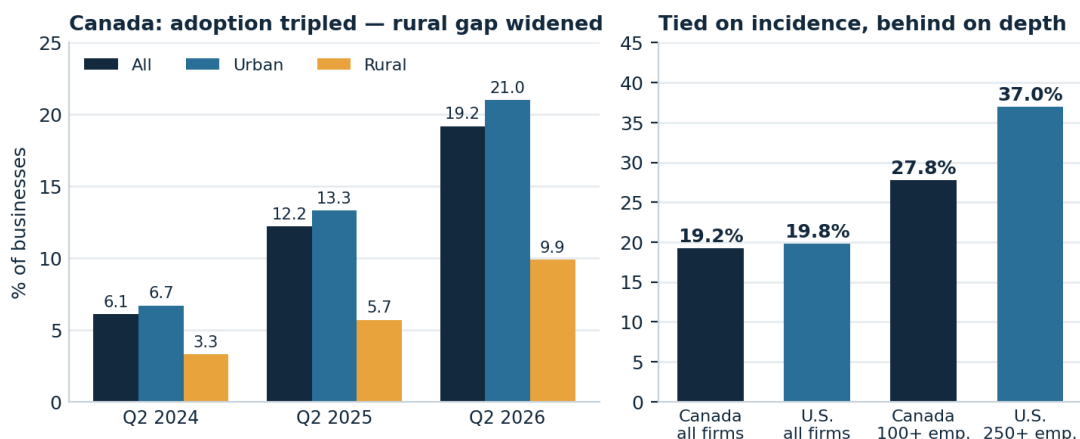
Six findings that should change an investment decision

- 1 Canada is tied with the U.S. on incidence, and behind on depth.** 19.2% of Canadian businesses used AI to produce goods or deliver services in the 12 months to Q2 2026, triple the 6.1% of Q2 2024 (StatCan CSBC, n=9,251). U.S. business AI use ran at 17–20% over Dec 2025–May 2026, at 19.8% on 3 May 2026 (U.S. Census BTOS). The real gap is among large firms: 37% for U.S. firms with 250+ employees versus 27.8% for Canadian firms with 100+ — a lower threshold, which understates it.
- 2 The productivity premium is not AI. It is the capability stack underneath AI.** Canadian firm-level work linking the Survey of Digital Technology and Internet Use to administrative microdata finds AI adopters 16.8% more productive than non-adopters. Control for pre-adoption productivity: 10.2%. Add complementary capabilities — R&D, cloud, data analytics, robotics, ICT training: 5.1%, and no longer statistically significant. **Do not pay a premium for AI adoption. Pay for, or build, the complements.**
- 3 Individual productivity gains are well evidenced. Enterprise EBITDA impact is not.** 80% of respondents say AI improved their own productivity, but only 37% of organisations report any positive EBIT contribution and 6% attribute 5% or more of EBIT to AI — both essentially flat year-on-year despite rising spend (McKinsey, Aug 2026, n=1,719). An NBER survey of ~6,000 executives across four countries found nine in ten reporting no productivity or employment effect over three years.
- 4 In manufacturing, AI returns follow a J-curve that lands inside the hold period.** U.S. Census manufacturing data (2017 and 2021) show causal J-curve returns: industrial AI use raises work-in-progress inventory and robot capex and depresses productivity and profitability in the short run, before adopters outperform. Losses concentrate among *older* establishments — the mid-market PE target profile. Roughly one-third of the loss in older plants traces to abandoning structured production-management practice.
- 5 AI has already repriced labour-arbitrage services — ahead of demonstrated demand loss.** Short interest in Teleperformance rose from 3.8% to over 12% of free float by May 2026 against a 2.4% peer average. Concentrix trimmed FY2026 revenue guidance roughly 1% and the shares fell over 21% premarket. **The first-order AI risk in services is terminal value, not margin.**
- 6 Distribution has the widest gap between intent and execution — which is the alpha.** 93% of wholesale distributors call AI a strategic priority; 16% have deployed it across multiple business functions (Distribution Strategy Group, 2026, n=233 executives, 57% C-suite). Where data quality is strong, 81% are confident in AI returns; where it is poor, 18% are. Wholesale trade had the second-lowest AI use of any Canadian industry, at 7.9%.

How to read this report. Findings are labelled **Observed** (measured), **Reported** (company- or vendor-stated), **Calculated** (my arithmetic from cited inputs) or **Forecast** (projection, not evidence). Vendor case studies are identified as such and are never treated as independent evidence of ROI. Two of the three rigorous causal studies cited use 2017–2021 data and therefore measure *pre-generative* AI — flagged at each point of use.

1. The adoption landscape: wide, shallow, and concentrated in language work

Two national statistical agencies now measure business AI use directly, and they agree. The convergence is the most commercially useful fact in the dataset: a Canadian mid-market industrial business is not operating in a market structurally behind its U.S. peer group on AI incidence — it is operating in one where *nobody* has gone deep.



Sources: Statistics Canada CSBC Q2 2024/2025/2026; U.S. Census Bureau Business Trends and Outlook Survey, 14 Dec 2025–3 May 2026. Note the size thresholds differ (100+ vs 250+ employees), which understates the large-firm gap.

Depth is the constraint, not incidence

Among U.S. adopting firms, 57% use AI in three or fewer business functions — most commonly Sales and Marketing (52%), Strategy (45%) and IT (41%): commercial and administrative work, not production (U.S. Census Bureau, CES-WP-26-25, 2026). The OECD found only ~10% of G7 businesses using AI for *core business functions* in 2024, with 2–6% at high intensity.

Selected AI applications used, by age of business (% of Canadian AI adopters, Q2 2026)

	≤2 yrs	3–10 yrs	11–20 yrs	>20 yrs
Data analytics	45.5	33.7	27.0	46.4
Text analytics	41.2	32.5	26.2	42.5
Virtual agents / chatbots	17.9	29.4	25.0	31.4
Large language models	21.1	25.8	24.0	25.0
Machine learning	34.0	14.8	19.0	18.9
Decision-making systems	10.0	15.1	15.7	11.2
Deep learning	13.3	14.7	13.1	10.9
Image / pattern recognition	17.9	14.4	11.3	14.9
Robotic process automation	0.0	4.2	4.2	7.6
Machine / computer vision	0.1	6.4	2.3	5.2

Industrial AI

Source: Statistics Canada, CSBC Q2 2026 (Table 33-10-1167-01). Base = the 19.2% of businesses using AI. Ten of eighteen reported applications shown.

Calculated. Machine or computer vision is used by 4.6% of Canadian AI adopters and robotic process automation by 5.0%. Applied to the 19.2% adoption base, that implies roughly **0.9% and 1.0% of all Canadian businesses** respectively. The applications that change industrial operating economics are, at national scale, rounding errors.

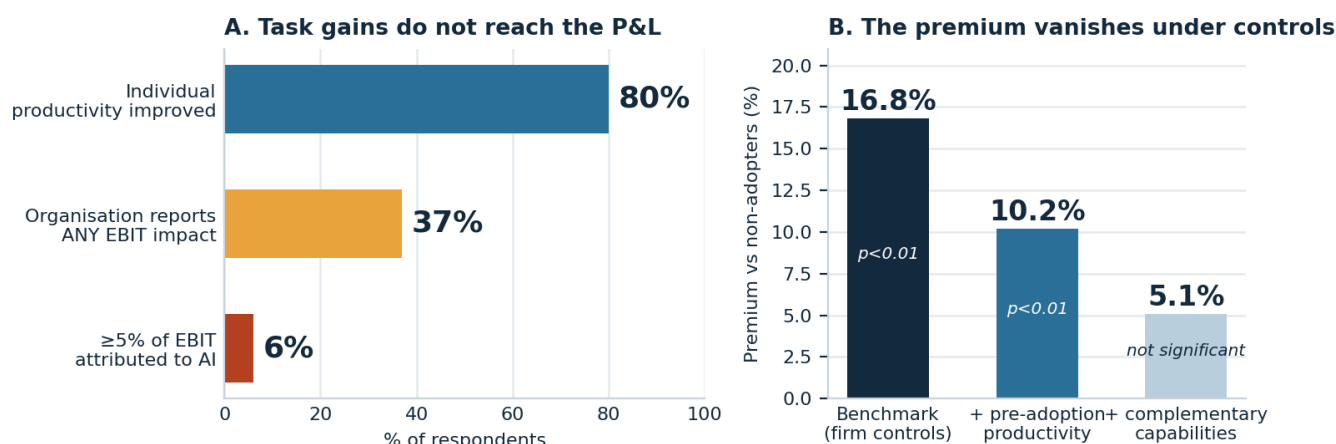
Observed, and directly relevant to deal flow. Businesses more than 20 years old adopt at 15.0% versus 25.0% for firms aged 3–10. The established, owner-operated industrial company in a typical Canadian mid-market pipeline sits in the *lowest-adopting cohort by age* — simultaneously the opportunity and the execution risk.

The binding barrier is capability, not cost or regulation. 78.1% of non-adopters cite irrelevance to their products, versus 13.4% citing cyber/privacy and 10.6% cost — but 12.0% of manufacturers cite lack of skilled workers, and manufacturing and wholesale trade are *less* likely than average to call AI irrelevant. They see it; they are blocked by capability.

Sources: Statistics Canada, CSBC Q2 2026 (Table 33-10-1167-01); U.S. Census Bureau BTOS; Bonney et al., CES-WP-26-25 (2026); Wyonch, C.D. Howe Commentary 712 (Apr 2026); Filippucci et al., OECD (2025).

2. AI economics: what is demonstrated, and what is only potential

The gap between task-level gain and enterprise financial impact is the central analytical problem for an investor. Both sides of it are well measured, and they do not meet.



A: McKinsey, The State of AI in 2026 (Aug 2026; n=1,719 across 97 countries; fielded 4 May–8 Jun 2026). Both the 37% and the 6% are essentially flat vs 2025. B: Li & Liu (forthcoming, Canadian Public Policy), reproduced in Statistics Canada, Economic and Social Reports, 22 Apr 2026. Data: Survey of Digital Technology and Internet Use, 2019 and 2021 waves linked to administrative microdata — i.e. PRE-generative AI. Complements = R&D, cloud, data analytics, robotics, ICT training.

Transmission mechanisms, graded by evidence

Mechanism	What the evidence shows	Status
Productivity	~15% more issues resolved per hour among 5,172 customer-support agents (Brynjolfsson, Li & Raymond, QJE, May 2025); novices +34%, top performers minimal gain and slight quality decline. ~17% average across GenAI studies in meta-analysis.	Demonstrated at task level
Revenue	Accenture FY2025: advanced AI revenue tripled to \$2.7bn on \$69.7bn total (3.9% of revenue); GenAI bookings nearly doubled to \$5.9bn. GAAP operating margin fell 10bps to 14.7% . Revenue gains most often attributed to marketing and sales.	Demonstrated for AI vendors; potential for AI users
Cost	Cost reductions most frequently reported in supply chain management, service operations and manufacturing (McKinsey 2026) — but this is respondent perception, not audited cost data.	Directionally supported; not quantified
Working capital	Evidence points the <i>opposite</i> way to the marketing: industrial AI adoption was found to <i>increase</i> work-in-progress inventory during adjustment. Forecasting-improves-inventory logic is sound in distribution, but no independent quantified evidence located.	Documented adverse effect in manufacturing during implementation
Asset utilisation	Predictive maintenance logic is coherent, but every quantified figure located (30–50% fewer stoppages, 20–40% longer asset life) traces to vendor material. Siemens does not disclose plant-level financial savings across its footprint.	Mechanism sound; independent evidence absent
EBITDA	37% of organisations report any positive EBIT contribution; 6% attribute ≥5%. Nine in ten executives in the NBER sample report no effect over three years.	Demonstrated for a minority ; not for the median firm

The one finding an investor should actually operationalise. The U.S. Census Bureau paper finds a robust positive correlation between commercial performance and the *breadth* of AI integration across functions, tasks and operational investment. McKinsey's high performers show the same pattern from a different dataset: roughly three-quarters had fundamentally redesigned workflows, versus one-quarter of everyone else. Both are cross-sectional and cannot establish direction of causation — so treat breadth of integration as a **screening indicator of management quality**, not as a lever that can be pulled on its own.

A benchmark for sizing claims. Sysco — \$84.6bn of FY2026 sales, 333 distribution centres — announced a three-year programme across routing, warehouse productivity, merchandising/procurement and back office targeting at least \$500m of net cost-out by FY2029 (8-K, 9 Sep 2026). *[Calculated]* That is about **0.6% of sales**. Any mid-market target claiming materially more should be asked to reconcile against it.

What remains unknown. There is no rigorous firm-level causal evidence on generative AI's effect on EBITDA — the two best causal studies use 2017–2021 data. No independent evidence quantifies AI's effect on technician productivity in industrial services, or on inventory turns in distribution. And the durability of AI pricing gains is untested. These are the four places where a thesis is currently unfalsifiable, and therefore the four places to demand target-specific data.

3. Sector view: where the economics actually work

Sector attractiveness map — every row oriented so HIGH = more attractive to an investor

	B2B Services	Manufacturing	Industrial Services	Distribution
Current AI adoption depth	HIGH	MOD	LOW	LOW
Independent evidence quality	HIGH	MOD	LOW	LOW
Near-term EBITDA impact (0-24 mo)	MOD	LOW	MOD	HIGH
Revenue-model opportunity	MOD	LOW	HIGH	LOW
Ease of implementation	HIGH	LOW	MOD	MOD
Working-capital opportunity	LOW	LOW	LOW	HIGH
Moat / differentiation potential	LOW	HIGH	HIGH	MOD
Business-model durability	LOW	HIGH	HIGH	MOD
PE value-creation relevance	MOD	MOD	HIGH	HIGH

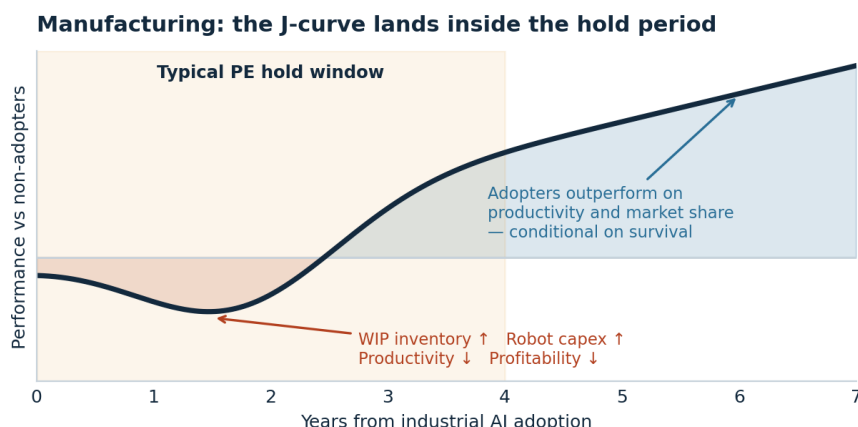
Methodology: three-level qualitative assessment (Low / Moderate / High) derived from the evidence cited in this report — not a numerical score, index or ranking. Two dimensions are stated in positive polarity so all rows read the same way: 'Ease of implementation' inverts implementation difficulty; 'Business-model durability' inverts disruption risk. No column totals are computed, because the rows are not commensurable and should not be summed.

Evidence base by sector

Sector	The mechanism that matters	Best company evidence	The investment call
B2B Services	AI raises the floor more than the ceiling — tacit senior expertise is partially transferred into the system. Margin opportunity and moat erosion at the same time.	Accenture (10-K/DEF 14A): AI revenue tripled, margin flat. Teleperformance / Concentrix : multiple compression ahead of revenue loss. TELUS Digital (6-K, Oct 2025): taken private at US\$4.50/sh, 52% premium to unaffected; EQT realised ~4x from a 2016 entry.	Selective. Underwrite the exit multiple conservatively, independent of EBITDA growth. Avoid time-and-materials fee pools.
Manufacturing	J-curve. Short-run productivity and profitability losses, rising WIP and robot capex, before adopters outperform on productivity and market share.	Vendor-dominated. Schneider Electric / Compass Datacenters : 40% fewer on-site interventions, 20% lower opex — <i>company-reported</i> . The widely cited Magna “287% ROI” figure has no traceable primary disclosure and is excluded .	Buy digital maturity at a premium, or budget the dip explicitly. Verify sensor coverage, MES and maintenance-record quality <i>before</i> evaluating any AI roadmap.
Industrial Services	Two chains: scheduling → travel time down → revenue per technician up; and asset monitoring → break-fix converted to contracted recurring revenue. The second changes the business model.	Precision Drilling (Calgary; 6-K Mar 2026): Alpha automation/analytics platform; AlphaArms robotics automating >95% of rig-floor activities on its first Super Triple rig. Financial contribution not separately disclosed . External ROI evidence is vendor-only.	Highest risk-adjusted . The metrics — utilisation, first-time fix, travel time — are measurable inside the target. Test vendor claims against its own dispatch data.
Distribution	Pricing is where the money is: thin gross margins mean price changes flow to EBITDA almost without friction, and the effect lands in the current quarter.	Sysco (8-K, Sep 2026): ≥\$500m AI cost-out by FY2029, ~\$100m in FY2027 guidance — <i>target, zero realised</i> . Grainger Q2 2026: +120bps operating margin attributed to mix, freight, price, UK exit and tariff refunds — not to AI .	Fastest EBITDA conversion . But treat the first-year pricing uplift as a one-time stock, not a growing flow , until proven otherwise.

When AI actually creates differentiation — three conditions, and only three. (1) **Proprietary data the target owns** and a competitor cannot buy: years of labelled defect, failure or service-history records. (2) **Multi-function integration that compounds**, where forecasting improves inventory, inventory improves service information, and service interactions improve forecasting. (3) **A workflow redesign that is organisationally hard to copy** — it requires change management, not procurement. Where none holds, AI is a cost of staying in business and the benefit is competed away to customers through price.

4. The private-equity framework



Schematic illustration of the pattern documented in McElheran, Yang, Kroff & Brynjolfsson, 'The Rise of Industrial AI in America', U.S. Census Bureau / AEA 2026, using Census manufacturing data for 2017 and 2021. The curve is drawn to convey shape and timing only — it is not a reproduction of their estimates, and the authors publish no year-by-year magnitudes. Losses concentrate among older establishments; young firms fare better.

Diligence: the questions that discriminate

Management claim	The question to ask	What a good answer contains
"AI will cut costs 20%"	Against what baseline, over what period, net of licence, integration and change-management cost?	A cost bridge by line item, reconciled to the GL. Sysco's benchmark is ~0.6% of sales over three years.
"AI creates a moat"	What is the input a competitor cannot buy — and do your OEM and customer contracts confirm you own it?	Proprietary, labelled, longitudinal operating data. Models and software are purchasable; data is not.
"We are AI-enabled"	In how many business functions is AI in daily production use, not pilot?	Count functions in production. 57% of U.S. adopters use AI in three or fewer; 16% of distributors are multi-function.
"AI improved our margin"	How did you separate AI from price, mix, freight, tariffs and operating leverage?	A full margin bridge. If the target cannot decompose it the way Grainger does, the attribution is unsupported.
"This lets us cut headcount"	Has anyone actually done it?	Almost nobody has. 2% of U.S. firms report AI-related employment decreases; 89.4% of Canadian adopters reported no employment change.

Value creation: sequence by payback, not sophistication

- 1. Contract and rebate extraction** (distribution) — cheapest; can pay for itself on a single recovered volume tier.
- 2. Pricing governance + AI price optimisation** — lands in the current fiscal year.
- 3. Scheduling, routing and dispatch** — 6–12 months; verifiable in the target's own data.
- 4. Master-data remediation** — no direct return, and therefore always deferred. Items 1, 2, 3 and 5 all depend on it. Budget it in the first 100 days.
- 5. Predictive maintenance / vision QC** — multi-year; expect a J-curve.

Exit: do not assume AI raises the multiple

It **can raise it** where AI has converted transactional revenue into contracted recurring revenue. It **can lower it** where a business reads as *AI-exposed* rather than *AI-enabled*. Most often it is **neutral**.

Bain and StepStone's 2026 GP Outlook found GPs report their highest generative-AI returns in **deal sourcing and due diligence** — their own workflow — while portfolio-company benefits skew to cost savings and **nearly 40% of GPs expect no material financial impact in 2026**. The buyer base is currently sceptical. **Underwrite the operational improvement, not the AI narrative.**

Principal sources

Tier 1 — statistical agencies, filings and academic work: Statistics Canada, *Analysis on AI use by businesses in Canada, Q2 2026* (11 Jun 2026, Table 33-10-1167-01) · Statistics Canada / ISED, *AI adoption and productivity in Canadian firms* (22 Apr 2026), summarising Li & Liu, forthcoming, *Canadian Public Policy* · U.S. Census Bureau, *Business Trends and Outlook Survey* (Dec 2025–May 2026) and Bonney et al., CES-WP-26-25 (Apr 2026) · Brynjolfsson, Li & Raymond, *QJE* 140(2) (2025) · McElheran, Yang, Kroff & Brynjolfsson, U.S. Census Bureau / AEA (2026) · Yotzov, Barrero, Bloom et al., NBER WP 34836 (2026) · OECD AI Papers No. 41 (2025) · company filings: Accenture DEF 14A FY2026, Sysco 8-K (9 Sep 2026), W.W. Grainger Q2 2026, TELUS 6-K (31 Oct 2025), Precision Drilling 6-K (Mar 2026).

Tier 2–3: Wyonch, C.D. Howe Commentary 712 (Apr 2026) · McKinsey, *The State of AI in 2026* (Aug 2026) · Bain & Co. / StepStone, *2026 PE GP Outlook and Global Private Equity Report 2026* · Distribution Strategy Group, *The AI Execution Gap* (Sep 2026) · CFIB; ITIF.

Excluded: the widely circulated Magna "287% ROI" figure, for which no primary disclosure could be located, and predictive-maintenance statistics published without traceable methodology.